

Disaster Resource Allocation & Command Center

1. Problem Statement

Natural disasters such as floods, earthquakes, cyclones, landslides, and fires often result in delayed rescue operations due to poor coordination, inefficient resource allocation, and lack of real-time communication. Existing systems mainly focus on reporting incidents but provide limited support for managing rescue resources, volunteers, shelters, and relief operations. This leads to duplicated efforts, resource shortages, and slower emergency response.

2. Objective

The project aims to develop a centralized web-based platform that helps disaster response teams efficiently manage incidents, allocate resources, coordinate volunteers, monitor rescue operations, and generate analytical reports for better decision-making.

3. Proposed Solution

The proposed **Disaster Resource Allocation & Command Center** is a web application developed using **React.js, Spring Boot, and PostgreSQL**. Control Room Operators register disaster incidents received through emergency helplines or local authorities. Based on disaster severity and resource availability, the system enables administrators to allocate rescue teams, ambulances, shelters, food supplies, medical kits, and volunteers.

The platform also provides real-time monitoring dashboards, interactive maps for disaster locations, and instant notifications to NGOs and volunteers using Firebase Cloud Messaging. Google Maps integration helps visualize disaster areas and nearby shelters, while Twilio SMS (optional) can send emergency alerts when internet connectivity is limited.

4. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Spring Boot
Database	PostgreSQL
Security	Spring Security + JWT
ORM	Spring Data JPA
Maps	Google Maps API
Notifications	Firebase Cloud Messaging
SMS Alerts	Twilio API (Optional)
API Testing	Postman
Version Control	Git & GitHub

5. User Roles

Administrator

- Manage users
- Allocate resources
- Monitor disaster operations
- Generate reports

Control Room Operator

- Register disaster incidents
- Update incident status
- Assign rescue teams

NGO Coordinator

- Manage relief resources
- Update inventory
- Track resource distribution

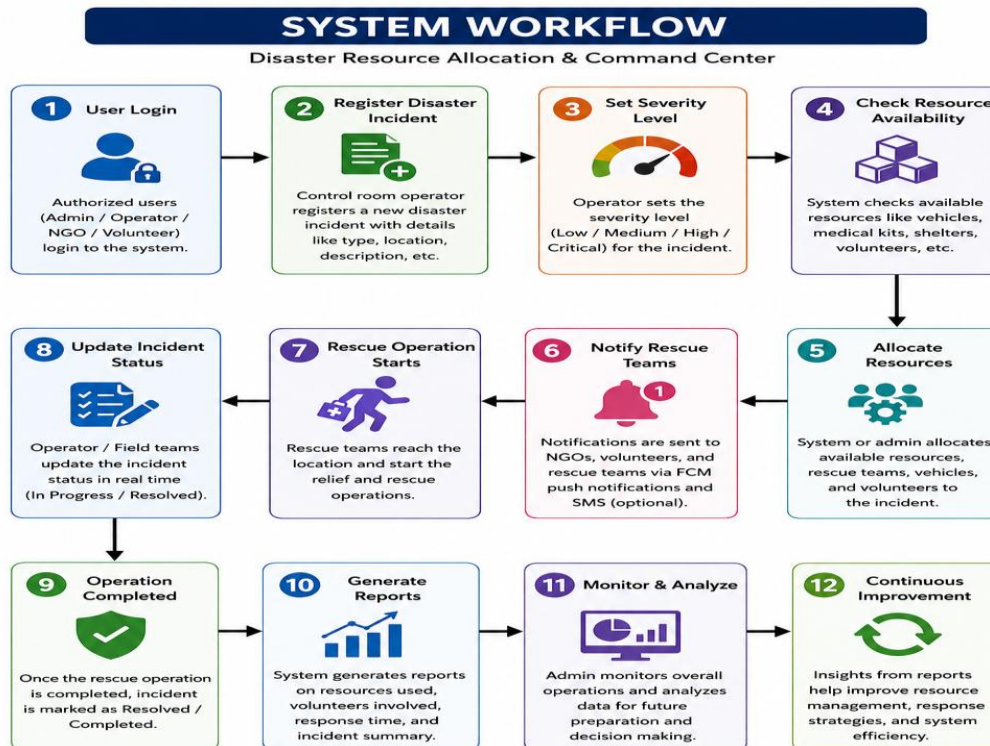
Volunteer

- View assigned tasks
- Update task progress
- Mark tasks as completed

6. Functional Modules

- **Authentication** – Secure login using JWT and role-based access.
- **Disaster Incident Management** – Register and manage disaster incidents.
- **Resource Management** – Manage food, medical kits, rescue vehicles, shelters, and emergency supplies.
- **Resource Allocation** – Assign available resources based on disaster severity.
- **Volunteer Management** – Register volunteers and assign rescue tasks.
- **Shelter Management** – Track shelter capacity and occupancy.
- **Notification System** – Send real-time alerts and updates.
- **Map & Location Tracking** – Display disaster locations and nearby shelters.
- **Reports & Dashboard** – View disaster statistics, resource utilization, and rescue progress.

7. System Workflow



8. Database Design

The system consists of the following main entities:

- **User** (Authentication & Roles)
- **Disaster** (Incident Details)
- **Resource** (Relief Materials & Rescue Equipment)
- **Allocation** (Resource Assignment)
- **Volunteer** (Volunteer Details & Availability)
- **Shelter** (Capacity & Occupancy)
- **Notification** (Alerts & Messages)

9. External APIs

- **Google Maps API** – Displays disaster locations, shelters, and resource locations.
- **Firebase Cloud Messaging (FCM)** – Sends real-time notifications to volunteers and NGOs.
- **Twilio SMS API (Optional)** – Sends emergency SMS alerts when internet connectivity is unavailable.

10. Expected Outcome

The proposed system provides a centralized platform for disaster response agencies to manage incidents, allocate resources, coordinate volunteers, monitor rescue operations, and generate reports efficiently. By integrating maps, notifications, and role-based management, the application improves communication, reduces response time, and enhances overall disaster relief coordination. The project demonstrates modern full-stack development using **React.js, Spring Boot, PostgreSQL, JWT Authentication, REST APIs, and third-party integrations**, making it suitable for final-year academic projects and software engineering portfolios.